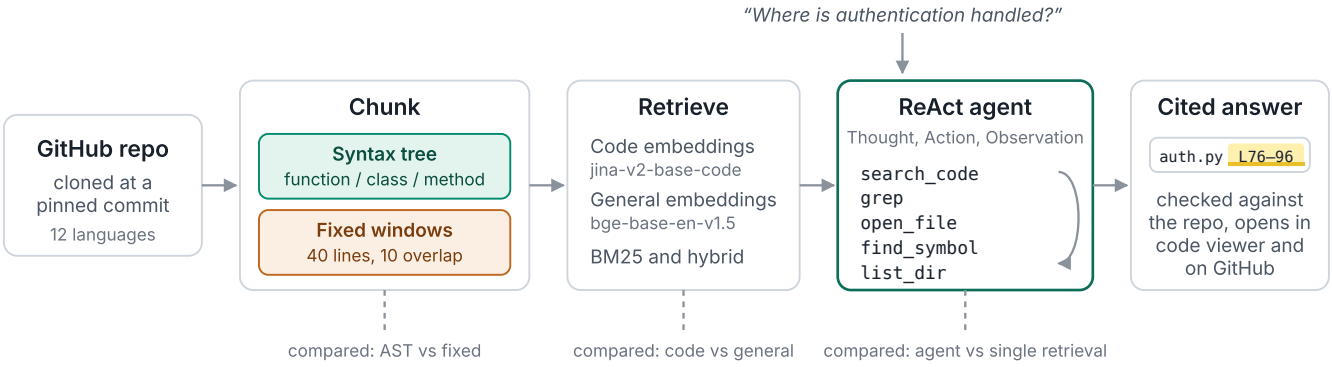
Codebase QA — Project Report

7 November 2026

We built an assistant that answers questions about a GitHub repository, such as *"where is authentication handled?"* or *"explain this function"*, and links every answer to the exact file and lines. A **ReAct agent** searches, greps and opens files in a repository pinned to one commit, then returns a cited answer. Each citation opens the code with those lines highlighted, in the app or as a GitHub permalink.

How it works



The syntax-tree chunker uses Python's `ast` module and tree-sitter for 11 other languages, keeping each function, class or method whole; large classes are split into a header plus one chunk per method. The baseline cuts fixed 40-line windows. Both are embedded with a code model and a general one of similar size, beside BM25 and a hybrid. The agent uses a plain-text ReAct protocol, so it runs on Gemini, Groq, OpenAI, Anthropic or a local Ollama model, and its steps stream live to the web app.

Evaluation and limitations

The test set has 50 questions over three repositories pinned to a release: 26 ask where something happens, 17 how it works and 7 ask to explain a function. Gold answers are written as a file and symbol, then resolved to exact lines with the syntax tree.

Repository	Version	Questions	Files indexed	Function chunks (avg lines)	Fixed windows (avg lines)
psf/requests	2.32.3	17	19	275 (19.7)	196 (38.5)
pallets/flask	3.0.3	17	24	359 (24.0)	304 (38.9)
encode/starlette	0.38.2	16	35	430 (15.0)	237 (37.8)
Total		**50**	**78**	**1,064**	**737**

We score all 8 indexes (2 chunkers × 4 retrievers) on **Recall@1, 3, 5 and 10**, MRR and top-1 file accuracy. A chunk counts as correct when it overlaps at least half of the expected span, or half of itself if it is the smaller one. Single-retrieval RAG and the agent then answer every question; an **LLM judge** scores each answer from 1 to 5 against a reference answer, and we check **file- and line-location accuracy** of its citations. 81 tests run offline, and one script, `run_demo.sh`, sets everything up and opens the web app.

Limitations. All three benchmark repositories are Python, so the other languages are tested but not scored. With 50 questions, one question moves a metric by 2 points, and the judge is itself an LLM, kept grounded by the reference answer and gold lines. Next, we will record the Recall@k and judge results with the real models and add a non-Python repository.